

1. AI-Introduction

Artificial Intelligence

Definition

Artificial Intelligence enables machines to perform tasks requiring human intelligence.

Applications

- Chatbots
- Self-driving Cars
- Healthcare
- Finance

2. Machine-Learning

Types

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Applications

- Spam Detection
- Recommendation Systems
- Fraud Detection

3. Deep-Learning

Definition

Deep Learning uses Artificial Neural Networks with multiple hidden layers.

Applications

- Image Recognition
- Speech Recognition
- NLP

4. LLM

Large Language Models

Examples

- ChatGPT
- Gemini
- Claude
- Llama

Uses

- Text Generation
- Code Generation
- Translation
- Summarization

5. AI-Applications

Applications of AI

Industries

- Healthcare
- Banking
- Education
- Manufacturing
- Agriculture
- Cybersecurity

Benefits

- Automation
- Improved Decision Making
- Higher Productivity

6. Prompt-Engineering

Definition

Prompt Engineering is the process of designing effective prompts to obtain accurate and useful responses from AI models.

Best Practices

- Be Specific
- Provide Context
- Use Examples
- Break Complex Tasks
- Iterate and Refine

Applications

- Content Generation
- Coding Assistance
- Data Analysis
- Learning Support